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Institute of Science and Technology

A CASE STUDY REPORT

ON

E-GOVERNANCE IMPLEMENTATION SCENARIO AND STATUS IN NEPAL

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A CASE STUDY REPORT SUBMITTED TO THE DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE DEGREE OF BACHELOR IN SCIENCE IN COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

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Executive Summary

This case study report examines the architecture, institutional governance, service maturity, and structural challenges of electronic governance (e governance) in the Federal Democratic Republic of Nepal. The analysis is situated within the theoretical framework of the United Nations E Government Development Index (EGDI), which evaluates national digital governance capacity along three composite pillars: the Online Service Index, the Telecommunications Infrastructure Index, and the Human Capital Index. In the E Government Survey 2024, Nepal was placed 119th among 193 member states with a composite EGDI score of 0.5781, an improvement of six positions from its 2022 ranking of 125th, yet the country continues to trail most South Asian peers with the exception of Pakistan and Afghanistan, and its E Participation Index ranking of 152nd underscores a persistent weakness in citizen facing digital interaction.

The report traces the historical evolution of information and communication technology (ICT) adoption in Nepali public administration, beginning with the establishment of the erstwhile Electronics Data Processing Center and continuing through the enactment of the Electronic Transactions Act 2063 (2006), the Right to Information Act 2064 (2007), and the more recent Individual Privacy Act 2075 (2018). It evaluates the Digital Nepal Framework (DNF) 2019, the country's principal policy blueprint, across its eight priority sectors, and assesses the institutional mandates of the Ministry of Communication and Information Technology (MoCIT), the Department of Information Technology (DoIT), the National Information Technology Center (NITC), and the Nepal Telecommunications Authority (NTA).

At the infrastructure layer, the study reviews the Government Integrated Data Center (GIDC) and its disaster recovery counterpart in Hetauda, the optical fiber backbone financed through the Rural Telecommunication Development Fund (RTDF), the national Public Key Infrastructure (PKI) regime for digital signatures, and the still incomplete Nepal Government Enterprise Architecture (NGEA) and Nepal e Governance Interoperability Framework (NeGIF). At the service layer, the report analyzes flagship platforms including the Nagarik App, the biometric National ID (Rastriya Parichayapatra), the Integrated Tax System of the Inland Revenue Department, ASYCUDA World customs clearance, the Nepal National Single Window, the e Passport system, the electronic Government Procurement (e GP) portal, and land administration platforms such as Mero Kitta.

The findings indicate that Nepal's e governance ecosystem has transitioned from an early, largely informational web presence toward a partially transactional model, but this transition remains uneven: centralized platforms suffer recurrent downtime during high demand periods, departmental databases remain siloed due to the absence of an enforced interoperability standard, and the digital divide between Bagmati Province and the Karnali and Sudurpashchim provinces continues to widen. Cybersecurity governance is nascent, with no fully operational, centralized Security Operations Center (SOC) for government domains despite recurring distributed denial of service (DDoS) incidents and website defacements. The report concludes that sustained political commitment, the enactment of the pending Information Technology and Cybersecurity Bill, enforceable interoperability mandates, and structured human capital retention strategies constitute the critical determinants of Nepal's future digital governance trajectory, and it closes with a prioritized, stakeholder differentiated set of recommendations spanning short, medium, and long term implementation horizons.

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CHAPTER 1: INTRODUCTION

Electronic governance, commonly abbreviated as e governance, denotes the application of information and communication technology by public sector institutions to deliver services, disseminate information, facilitate citizen participation, and manage internal administrative processes. Contemporary digital governance scholarship, including the theoretical model underlying the United Nations E Government Development Index (EGDI), decomposes national digital governance capacity into three interdependent dimensions: the Online Service Index (the sophistication and transactional depth of government web and mobile platforms), the Telecommunications Infrastructure Index (broadband penetration, internet subscriber density, and mobile cellular coverage), and the Human Capital Index (adult literacy, gross enrolment ratios, and expected years of schooling). A fourth complementary construct, the E Participation Index, measures the extent to which digital platforms enable citizens to access information, engage in consultation, and participate in decision making. Nepal's digital governance trajectory must be assessed against this composite framework rather than through isolated project level observations, since strong performance in a single flagship application, such as a mobile citizen services app, does not by itself indicate mature digital governance if the underlying telecommunications infrastructure, legal architecture, and institutional interoperability remain weak.

1.1 Historical Evolution of ICT in Nepal's Public Administration

The institutional history of computing in the Nepali civil service dates to 1971, when the Electronics Data Processing Center (subsequently renamed the National Computer Center) was established under the National Planning Commission to computerize national census processing. Through the 1980s and 1990s, ICT adoption remained confined to isolated data processing functions within specific ministries, with no unifying legal or architectural framework. The turning point in the legal foundation of Nepali e governance arrived with the enactment of the Electronic Transactions Act 2063 (2006), which conferred legal recognition upon electronic records and digital signatures, established the offence of cybercrime under Nepali jurisprudence for the first time, and created the institutional basis for a national Certifying Authority to issue digital certificates. This was followed by the Right to Information Act 2064 (2007), which imposed a statutory obligation on public bodies to proactively disclose information and respond to citizen

information requests, indirectly accelerating the publication of government data through web portals. The formulation of the National e Government Master Plan (2007) under the Ministry of Science and Technology, financed with technical assistance from the Government of Korea, marked the first attempt at a coordinated, multi year digital government roadmap, though implementation lagged significantly behind the plan's projected timelines due to resource constraints and repeated changes in the lead ministry's mandate. The most consequential recent milestone is the promulgation of the Digital Nepal Framework in 2019, discussed at length in Chapter 2, which for the first time positioned digital transformation as a cross sectoral national development priority rather than a narrow ICT ministry initiative.

1.2 Scope of the Report

This report examines digital governance implementation across three tiers of Nepal's federal structure established under the Constitution of Nepal 2072 (2015): the federal government (encompassing ministries, departments, and centrally operated platforms such as the Nagarik App and GIDC), the seven provincial governments, and the 753 local governments (metropolitan cities, sub metropolitan cities, municipalities, and rural municipalities). The analysis covers policy instruments, legal statutes, core technical infrastructure, and citizen and business facing digital services operated by civilian public authorities. Classified defense, intelligence, and national security communication networks, along with proprietary systems of state owned enterprises operating under separate corporate governance, fall outside the scope of this report, consistent with the publicly verifiable, non classified nature of the source material relied upon.

1.3 Methodology

The report adopts a desk based, qualitative case study methodology. Primary source material includes government policy documents (the Digital Nepal Framework 2019, sectoral ICT policies, and the Fifteenth and Sixteenth periodic development plans), statutory texts (the Electronic Transactions Act, the Right to Information Act, and the Individual Privacy Act), operational data and annual reports published by the National Information Technology Center and the Nepal Telecommunications Authority, and international benchmarking instruments, principally the United Nations E Government Survey series. Secondary sources comprise peer reviewed literature on e governance in developing and least developed country contexts, independent audits and

evaluations of Nepali digital government platforms, and contemporaneous reporting from established Nepali and international technology and policy outlets. Findings are triangulated across at least two independent sources wherever quantitative claims are advanced, and the report explicitly flags areas where authoritative, up to date data is unavailable rather than extrapolating unsupported figures.

1.4 Research Questions and Problem Definition

The report is organized around the following research questions.

- To what extent has Nepal's legal and institutional architecture kept pace with the technical requirements of transactional, interoperable digital government?
- What is the operational maturity of core shared infrastructure, namely the Government Integrated Data Center, the national fiber backbone, and the Public Key Infrastructure regime?
- How mature are Nepal's principal citizen facing and enterprise facing digital services, measured against Government to Citizen (G2C), Government to Business (G2B), and Government to Government (G2G) service maturity criteria?
- What structural barriers, including data silos, vendor lock in, procurement inefficiency, and the rural urban digital divide, constrain further progress?
- What differentiated policy and technical interventions would most effectively accelerate Nepal's digital governance maturity across the federal, provincial, and local tiers?

The problem definition underlying this study is that Nepal's digital governance ecosystem exhibits a persistent gap between policy ambition, as articulated in instruments such as the Digital Nepal Framework, and operational execution, as evidenced by fragmented departmental systems, an EGDI ranking that remains in the lower half of the global distribution, and recurring service reliability failures during periods of peak citizen demand.

CHAPTER 2: POLICY, LEGAL, AND INSTITUTIONAL ARCHITECTURE

Effective e governance rests upon a coherent legal foundation that confers evidentiary status on electronic records, protects personal data, criminalizes unauthorized system access, and assigns unambiguous institutional mandates. This chapter reviews Nepal's legislative backbone, its principal policy blueprint, and the institutional landscape responsible for implementation.

2.1 Legislative Backbone

2.1.1 Electronic Transactions Act 2063 (2006)

The Electronic Transactions Act (ETA) remains the foundational statute governing digital transactions, electronic signatures, and cybercrime in Nepal. It establishes the legal equivalence of electronic and paper records for most administrative and commercial purposes, creates the Office of the Controller of Certification to license and supervise Certifying Authorities issuing digital certificates, and defines a set of computer related offences, including unauthorized access, damage to computer systems, and electronic fraud, with associated penalties. Successive amendments have attempted to extend the Act's coverage to newer categories of harm, including online defamation and social media misuse, but practitioners and academics have widely observed that the Act predates the proliferation of cloud computing, mobile applications, and cross border data flows, and consequently lacks provisions addressing data localization, cloud service provider liability, and algorithmic accountability.

2.1.2 Right to Information Act 2064 (2007)

The Right to Information Act obliges public bodies to designate information officers, maintain proactively disclosed categories of information, and respond to citizen requests within statutory timelines. While the Act does not mandate a specific digital publication format, it has functioned as an indirect driver of e governance by creating a legal incentive for ministries and departments to publish budgets, procurement notices, and organizational data on official websites.

2.1.3 Individual Privacy Act 2075 (2018)

The Individual Privacy Act represents Nepal's first comprehensive personal data protection statute, establishing principles for the collection, processing, and retention of personal information, and

creating obligations for public and private bodies that handle citizen data, including the biometric and demographic data underpinning the National ID system. The Act, however, does not establish an independent data protection authority with investigatory and enforcement powers comparable to those found in more mature regulatory regimes, and its enforcement mechanisms rely heavily on existing administrative and judicial channels, which limits its practical deterrent effect.

2.1.4 The Pending Information Technology and Cybersecurity Bill

Successive drafts of a consolidated Information Technology and Cybersecurity Bill, intended to replace or substantially amend the Electronic Transactions Act, have circulated within the federal parliament for several legislative sessions without enactment. The draft legislation reportedly addresses social media platform registration and liability, expanded cybercrime categories, data localization requirements for critical government data, and the establishment of a dedicated cybersecurity regulatory body. The prolonged legislative delay constitutes a material governance risk, since the absence of an updated statutory basis leaves emerging risks, including ransomware directed at government systems and cross border cloud hosting of citizen data, without a clear regulatory anchor.

2.2 Policy Blueprint: The Digital Nepal Framework (DNF) 2019

The Digital Nepal Framework, endorsed by the Government of Nepal in 2019 and subsequently updated, constitutes the country's principal strategic instrument for digital transformation. The Framework identifies a Digital Foundation layer, comprising shared infrastructure, interoperability standards, and digital identity, as a prerequisite for eight priority sectors: Digital Foundation, Agriculture, Health, Education, Energy, Tourism, Finance, and Urban Infrastructure. Each sector is associated with a defined set of digital initiatives intended to be delivered within specified timeframes. In practice, implementation has been markedly uneven across sectors. The Digital Foundation and Finance sectors have registered the most visible progress, evidenced by the rollout of the Nagarik App and the expansion of digital payment interoperability under the National Payment Switch operated by Nepal Clearing House Limited in coordination with Nepal Rastra Bank. Conversely, the Agriculture, Energy, and Urban Infrastructure sectors report comparatively limited delivery against DNF targets, reflecting weaker institutional ownership within the corresponding line ministries and the absence of ring fenced budgetary allocation tied specifically to DNF initiatives rather than to general ministry operating budgets.

2.3 Institutional Landscape and Jurisdictional Mandates

Nepal's e governance institutional architecture is characterized by a multiplicity of actors with overlapping or ambiguously delineated mandates. The Ministry of Communication and Information Technology (MoCIT) holds overall policy formulation authority for the ICT sector, including telecommunications regulation policy and the oversight of the Digital Nepal Framework's implementation. The Department of Information Technology (DoIT), operating under MoCIT, is responsible for technical standard setting, software quality assurance for government systems, and the licensing of certain ICT service providers. The National Information Technology Center (NITC), also under MoCIT, operates the Government Integrated Data Center, provides shared hosting and email services to government agencies, and functions as the de facto technical implementation arm for cross government infrastructure. The High Level e Governance Board, chaired at the level of the Prime Minister or a senior cabinet minister depending on the prevailing government structure, is intended to provide apex level coordination and dispute resolution across ministries, though its meeting frequency and effective authority have varied considerably across successive governments. The Nepal Telecommunications Authority (NTA), established under the Telecommunications Act 2053 (1997), regulates telecommunications service providers, manages spectrum allocation, and administers the Rural Telecommunication Development Fund, which finances last mile connectivity infrastructure in underserved districts.

Table 1: Institutional Governance Framework and Jurisdictional Mandates in Nepal

Institution	Parent Authority	Core Mandate	Key Limitation
Ministry of Communication and Information Technology (MoCIT)	Council of Ministers	ICT and e governance policy formulation; oversight of the Digital Nepal Framework	Frequent leadership turnover disrupts multi year policy continuity
Department of Information Technology (DoIT)	MoCIT	Technical standards, software quality assurance, ICT service provider licensing	Limited enforcement authority over non compliant line agencies
National Information Technology Center (NITC)	MoCIT	Operation of GIDC and DR site; shared hosting, email, and connectivity services	Capacity constraints relative to the scale of government hosting demand
High Level e Governance Board	Office of the Prime Minister and Council of Ministers	Apex coordination and cross ministerial dispute resolution	Irregular convening frequency; non binding recommendations in practice

Institution	Parent Authority	Core Mandate	Key Limitation
Nepal Telecommunications Authority (NTA)	MoCIT (regulatory autonomy)	Telecommunications licensing, spectrum management, Rural Telecommunication Development Fund	Fund disbursement delays for last mile rural connectivity projects
Department of National ID and Civil Registration (DoNIDCR)	Ministry of Home Affairs	National ID issuance, civil registration data management	Incomplete integration with sectoral databases (health, education, social security)

CHAPTER 3: OBJECTIVES OF THE STUDY

3.1 Primary Objectives

- To examine the legal, policy, and institutional architecture underpinning e governance implementation in Nepal, with particular reference to the Digital Nepal Framework 2019 and its constituent sectoral programs.
- To assess the operational status and technical maturity of Nepal's core shared digital infrastructure, including the Government Integrated Data Center, the national optical fiber backbone, and the Public Key Infrastructure regime.
- To evaluate the service maturity, adoption, and reliability of principal citizen facing and enterprise facing digital platforms across the Government to Citizen, Government to Business, and Government to Government service categories.
- To identify the structural, technical, and socio economic barriers that constrain the scaling and reliability of e governance services across Nepal's federal, provincial, and local tiers.

3.2 Secondary Objectives

- To audit the technical benchmarks (hosting tier, uptime, virtualization, and disaster recovery readiness) of the Government Integrated Data Center relative to accepted data center industry standards.
- To analyze the state of interoperability between departmental information systems and evaluate the practical adoption of the Nepal e Governance Interoperability Framework.
- To assess cybersecurity governance, incident response capacity, and the institutional readiness of Nepal's Computer Emergency Response Team (CERT) function.
- To formulate differentiated, actionable policy and technical recommendations addressed separately to the federal government, infrastructure operators, provincial and local governments, and academic and industry stakeholders.

CHAPTER 4: CORE INFRASTRUCTURE AND PLATFORM ANALYSIS

The reliability, scalability, and security of citizen facing digital services are fundamentally constrained by the maturity of the underlying shared infrastructure layer. This chapter examines four pillars of that layer: the centralized data center, the terrestrial fiber backbone, the digital signature and certification regime, and the enterprise architecture and interoperability standards intended to unify departmental systems.

4.1 Government Integrated Data Center (GIDC) and Disaster Recovery Site

The Government Integrated Data Center, operated by the National Information Technology Center in Singha Durbar, Kathmandu, functions as the principal shared hosting facility for federal government websites, databases, and select transactional applications, complemented by a geographically separated Disaster Recovery (DR) site located in Hetauda, Makwanpur district. The GIDC provides virtualized server hosting, shared storage, and managed connectivity to agencies that lack the technical capacity or budget to operate independent data centers, and its establishment represented a deliberate policy choice to consolidate fragmented, agency level server rooms into a centrally managed facility with improved physical security and standardized operational procedures. Notwithstanding this consolidation, the facility's hosting tier does not meet the redundancy criteria associated with internationally recognized Tier III or Tier IV data center classifications, principally due to constraints in redundant power feeds, cooling capacity, and carrier diversity. Power resilience remains a recurring operational concern, with the facility relying on diesel generator backup and uninterruptible power supply units whose runtime capacity is finite during Nepal's periodic extended load shedding episodes or generator maintenance windows. Cloud readiness, in the sense of elastic, on demand compute provisioning comparable to commercial hyperscale cloud platforms, remains limited; the GIDC's virtualization layer supports server consolidation and basic workload migration but does not yet offer a self service infrastructure as a service (IaaS) provisioning model to onboarded agencies. The DR site in Hetauda provides a secondary hosting location intended to support business continuity in the event of a primary site outage, but replication practices vary by hosted application, and not all critical government systems maintain real time or near real time data replication to the DR facility, which constrains the achievable Recovery Point Objective (RPO) for those systems.

4.2 Optical Fiber Backbone and the Information Superhighway

Nepal's terrestrial fiber optic backbone has expanded substantially over the past decade, driven by a combination of state owned carrier investment (principally through Nepal Telecom), private sector internet service providers, and the Rural Telecommunication Development Fund administered by the Nepal Telecommunications Authority. The Mid Hill Highway optical fiber expansion, financed in part through RTDF subsidies, extends terrestrial connectivity along the Mid Hill corridor traversing multiple districts, with the strategic objective of reducing dependence on costlier and less reliable microwave and satellite backhaul in hill and mountain districts. Despite this expansion, last mile connectivity in Karnali Province and large portions of Sudurpashchim Province remains disproportionately dependent on wireless backhaul, and fiber redundancy, meaning the availability of multiple physically diverse paths to a given locality, is limited outside the Kathmandu Valley and a small number of provincial capitals, which increases the risk of prolonged localized outages following fiber cuts caused by landslides, road construction, or other physical disruption.

4.3 Public Key Infrastructure (PKI) and Digital Signatures

The Office of the Controller of Certification, established under the Electronic Transactions Act, functions as the national root Certifying Authority responsible for licensing subordinate Certifying Authorities and issuing digital certificates that underpin legally recognized digital signatures. Notwithstanding this legal foundation, adoption of digital signatures within routine administrative workflows remains limited. Several structural factors constrain uptake: the absence of a mandatory digital signature requirement for the majority of internal government correspondence and approval workflows, limited integration of PKI based signing into the document management systems used by line ministries, and a persistent institutional preference for wet ink signatures on printed documents, particularly for financial approvals and inter agency correspondence, reflecting both procedural habit and residual uncertainty regarding the evidentiary treatment of digitally signed records in certain administrative and quasi judicial proceedings.

4.4 Nepal Government Enterprise Architecture (NGEA) and NeGIF

The Nepal Government Enterprise Architecture and the accompanying Nepal e Governance Interoperability Framework were conceived to establish common data exchange standards, shared

API gateways, and consistent architectural principles across government information systems, thereby enabling departmental applications, such as civil registration, tax administration, and land records, to exchange verified data without requiring citizens to resubmit the same documentation to multiple agencies. In practice, NeGIF adoption remains largely voluntary and inconsistently enforced; individual departments continue to procure and operate systems independently, frequently through separate donor funded or vendor led projects that specify proprietary data formats and authentication mechanisms not aligned with a common interoperability layer. The result is a proliferation of point to point integrations, where they exist at all, rather than a unified, API gateway mediated exchange architecture, which increases both the marginal cost of each new integration and the operational risk associated with maintaining numerous bespoke interfaces.

Table 2: Technical Benchmark and Operational Capacity of Government Core Infrastructure

Infrastructure Component	Operating Authority	Current Capability	Principal Constraint
Government Integrated Data Center (GIDC)	NITC	Virtualized shared hosting for federal agency websites and select transactional systems	Sub Tier III redundancy; limited elastic provisioning
Disaster Recovery Site, Hetauda	NITC	Secondary hosting location for business continuity	Inconsistent real time replication across hosted applications
National Fiber Backbone (Mid Hill corridor)	Nepal Telecom, private ISPs, RTDF financed segments	Expanding terrestrial connectivity along the Mid Hill Highway	Limited path redundancy outside Kathmandu Valley and provincial capitals
Public Key Infrastructure (Root CA)	Office of the Controller of Certification	Legal basis for digital signatures and certificate issuance	Low integration into routine administrative workflows
NGEA and NeGIF Interoperability Layer	DoIT and NITC	Defined data exchange standards and reference architecture	Voluntary adoption; limited enforcement across line agencies

CHAPTER 5: CITIZEN FACING AND ENTERPRISE E SERVICES IN NEPAL

This chapter surveys the principal digital platforms through which Nepali citizens and businesses interact with public authorities, spanning identity and civil registration, unified citizen services, revenue and fiscal administration, sector specific deployments, and local government digitalization.

5.1 Identity and Civil Registration Rails

The Department of National ID and Civil Registration (DoNIDCR), under the Ministry of Home Affairs, administers the national biometric identity program, issuing the Rastriya Parichayapatra (National Identity Card), which captures demographic data alongside biometric identifiers including fingerprints and iris scans. The National ID is designed to function as a foundational identity credential capable of underpinning eligibility verification for social security allowances, financial account opening, SIM card registration, and other services that require reliable identity assurance. Integration between the National ID database and dependent systems, including the Social Security Allowance disbursement system operated at the local government level, remains partial; many local governments continue to rely on legacy citizenship certificate numbers rather than the National ID number as the primary identity key, which constrains the realization of a single, authoritative identity record across service domains.

5.2 Unified Citizen Service Delivery: The Nagarik App

The Nagarik App, developed and maintained under the coordination of the Ministry of Communication and Information Technology in partnership with NITC and participating line agencies, functions as Nepal's principal unified mobile citizen services platform. Architecturally, the application operates as a front end aggregation layer that orchestrates calls to backend Application Programming Interfaces (APIs) exposed by individual departmental systems, rather than owning the underlying transactional data itself. Through this orchestration model, the application provides citizens with access to services including Permanent Account Number (PAN) registration status, driving license and vehicle related information linked to the Department of Transport Management, educational certificate verification against records held by relevant examination boards, voter roll status verification linked to the Election Commission, and a citizen

feedback and grievance submission channel. The aggregation architecture offers the advantage of allowing individual departments to retain ownership of their systems of record while presenting a consolidated interface to citizens, but it also means that the application's reliability for any given service is bounded by the reliability of the corresponding backend department system; when a departmental backend experiences downtime or performance degradation, the corresponding function within the Nagarik App becomes unavailable even though the application's own front end infrastructure may be functioning normally.

5.3 Revenue and Fiscal Systems

The Inland Revenue Department (IRD) operates the Integrated Tax System, a web based platform supporting Permanent Account Number registration, value added tax return filing, and income tax return filing, complemented by the electronic Tax Deduction at Source (e TDS) module used by withholding agents to report and reconcile deducted tax. Nepal's customs administration relies on ASYCUDA World, a customs management system developed under the United Nations Conference on Trade and Development (UNCTAD) program and deployed at Nepal's principal customs points, including the Tribhuvan International Airport Customs Office and major land border customs points such as Birgunj. The Nepal National Single Window (NNSW) initiative, intended to allow traders to submit standardized trade related documentation to multiple regulatory agencies through a single electronic gateway, has been implemented in phases, with continued integration work required to bring the full set of regulatory agencies involved in cross border trade, including quarantine, standards, and quality control authorities, onto the common platform.

5.4 Sectoral Digital Deployments

The Department of Passports has digitized passport application intake through an online booking and application system supporting the issuance of machine readable and electronic passports (e Passports), reducing, though not eliminating, the reliance on in person queuing for the initial application stage. The Department of Transport Management maintains a Driving License and Vehicle Registration Management Information System that continues to experience processing backlogs, particularly for driving license written and practical examination scheduling, reflecting capacity constraints relative to application volume rather than a purely technical system limitation. The Public Procurement Monitoring Office administers the electronic Government Procurement

(e GP) portal, through which public entities are required to publish procurement notices and, for many categories of procurement, receive electronic bid submissions, representing one of the more mature G2B transactional platforms in the Nepali government ICT landscape. Land administration has seen the introduction of the Property Assessment and Revenue Management System (PARMS) at land revenue offices and the Mero Kitta parcel information platform, which provides citizens with online access to select cadastral and parcel level information, though full transactional land registration continues to require in person attendance at land revenue offices in most jurisdictions.

5.5 Local Government Digitalization

At the local government tier, digitalization has progressed unevenly across the 753 local units. The Electronic Building Permit System (e BPS) has been adopted by a subset of urban and peri urban municipalities to digitize building permit application review, though rural municipalities with lower construction volumes and limited technical staff frequently continue to process building permits manually. Local Level Revenue Management Systems (LLRMS) support property tax assessment and business registration fee collection at the municipal level, but system adoption, data quality, and integration with the federal Financial Comptroller General Office's reporting infrastructure vary considerably between well resourced metropolitan cities and smaller rural municipalities, producing a pronounced disparity in digital service maturity between ward offices in Kathmandu Valley municipalities and those in remote hill and mountain local units.

Table 3: Status, Adoption Metrics, and Technology Stacks of Key Nepali E Services

Service Platform	Operating Agency	Service Category	Maturity Status
Nagarik App	MoCIT / NITC (API aggregation)	G2C	Operational; reliability bounded by backend department systems
National ID (Rastriya Parichayapatra)	DoNIDCR, Ministry of Home Affairs	G2C / G2G	Rollout ongoing; partial integration with dependent systems
Integrated Tax System and e TDS	Inland Revenue Department	G2B	Operational; core transactional filing functions in routine use
ASYCUDA World	Department of Customs	G2B	Operational at major customs points; phased NNSW integration
e GP Portal	Public Procurement Monitoring Office	G2B	Mature; mandatory publication and electronic bidding for many categories

Service Platform	Operating Agency	Service Category	Maturity Status
e Passport and Online Booking	Department of Passports	G2C	Operational; application intake digitized, issuance capacity constrained
DLRS (Driving License MIS)	Department of Transport Management	G2C	Operational with recurring scheduling and examination backlogs
Mero Kitta and PARMS	Land revenue offices	G2C	Partial; information access online, transactional registration largely in person
e BPS	Select municipalities	G2C	Adopted by a subset of urban local governments

Table 4: Maturity Matrix of Government to Citizen (G2C), Government to Business (G2B), and Government to Government (G2G) Services

Service Category	Representative Platforms	Predominant Maturity Level	Key Gap
Government to Citizen (G2C)	Nagarik App, e Passport, National ID, DLRS	Interactive to partially transactional	Inconsistent backend reliability and incomplete end to end digital completion
Government to Business (G2B)	e GP Portal, Integrated Tax System, ASYCUDA World, NNSW	Transactional	Fragmented integration across regulatory agencies in cross border trade
Government to Government (G2G)	NeGIF data exchange, inter departmental verification calls	Informational to emerging transactional	Absence of enforced, universal interoperability standard

CHAPTER 6: COMPARATIVE AND STRUCTURAL ANALYSIS

6.1 Centralized versus Decentralized Delivery Models

Nepal's shift to a federal structure under the 2015 Constitution created an inherent tension between the centralized hosting model embodied by the GIDC and the constitutionally guaranteed autonomy of provincial and local governments over their own digital service delivery choices. Centralization offers economies of scale in security management, standardized backup practices, and reduced duplication of infrastructure investment, but it also concentrates operational risk: a GIDC wide outage or capacity constraint simultaneously affects every federal agency hosted on the shared platform. Decentralization, exercised by local governments procuring independent software vendors for municipal service systems, increases resilience against any single point of failure but produces the fragmentation and inconsistent data standards documented in Chapter 5, since individual municipalities frequently select different vendors with incompatible data schemas and without a mandated common integration layer.

6.2 Service Reliability and Downtime Analysis

Empirical observation of Nepal's higher traffic government platforms reveals a recurring pattern in which service degradation or outright unavailability coincides with periods of concentrated citizen demand, most visibly in passport appointment booking windows following academic result publications or visa application seasons, and in driving license examination slot booking following periodic increases in application volume. This pattern is consistent with underlying infrastructure that has not been provisioned or architected for elastic scaling; systems designed around fixed, on premises virtual machine capacity at the GIDC lack the automatic horizontal scaling capabilities that commercial cloud native architectures provide during demand spikes, resulting in queueing, timeouts, and, in more severe cases, full service unavailability during peak intervals.

6.3 Interoperability Evaluation

The comparative evidence assembled in Chapters 2, 4, and 5 indicates that Nepal's e governance ecosystem remains predominantly organized around siloed departmental databases rather than a unified, API mediated data exchange layer. Where integration exists, such as the limited data flows supporting the Nagarik App's aggregation of departmental services, it has typically been achieved through bespoke, point to point technical arrangements negotiated individually between NITC or

the application development team and each participating department, rather than through universal adoption of the NeGIF reference architecture. This pattern increases both the marginal integration cost of onboarding each new service and the long term maintenance burden, since each point to point interface must be independently monitored, versioned, and secured.

6.4 Geographic and Socio economic Accessibility

Digital service accessibility in Nepal exhibits a pronounced geographic concentration favoring Bagmati Province, and particularly the Kathmandu Valley, where fiber connectivity density, smartphone penetration, digital literacy, and the physical presence of government technical staff are all comparatively higher than in other provinces. Karnali Province and large portions of Sudurpashchim Province, by contrast, continue to experience materially lower fixed and mobile broadband penetration, greater reliance on costlier data plans relative to household income, and a shortage of technically trained civil service personnel able to operate and maintain digital systems at the local level, all of which compound to produce a substantive digital exclusion gap that risks widening rather than narrowing as centrally designed digital services become more advanced without commensurate investment in provincial and rural digital readiness.

Table 5: Comprehensive Evaluation Across Five Strategic Criteria: Cost Efficiency, Service Reliability, Interoperability, Accessibility, and Security

Criterion	Current State	Principal Driver of Weakness	Directional Trend
Cost Efficiency	Moderate; shared GIDC hosting reduces duplicate agency infrastructure spend	Duplicative procurement at the local government tier	Improving, gradually
Service Reliability	Weak to moderate; recurring peak load outages on high traffic platforms	Fixed capacity, non elastic hosting architecture	Stagnant
Interoperability	Weak; predominantly siloed departmental databases	Voluntary, unenforced NeGIF adoption	Slowly improving
Accessibility	Uneven; strong in Bagmati Province, weak in Karnali and Sudurpashchim	Infrastructure and digital literacy disparities	Widening gap risk
Security	Weak; no fully centralized SOC for government domains	Absence of enacted cybersecurity legislation and dedicated CERT capacity	Improving, from a low base

CHAPTER 7: RISKS, CYBERSECURITY, AND OPERATIONAL CHALLENGES

7.1 Cyber Threat Landscape

Government domains under the .gov.np namespace have experienced recurring cybersecurity incidents, including distributed denial of service (DDoS) attacks that have periodically rendered ministry and departmental websites unreachable, website defacement incidents in which threat actors have altered publicly visible page content, and reported instances of unauthorized attempts to access or exfiltrate data from government held databases. These incidents illustrate the general vulnerability profile of a rapidly digitizing public sector that has not yet matched its expansion of citizen facing digital surface area with a commensurate investment in defensive security controls, monitoring, and incident response capability.

7.2 Lack of a Centralized Security Operations Center and Incident Response

Nepal established a national Computer Emergency Response Team (Nepal CERT related coordination function) to serve as a central point for cybersecurity incident coordination, but the country lacks a fully resourced, continuously staffed, centralized Security Operations Center with real time monitoring authority across all government hosted systems. Patch management, in the sense of the systematic, timely application of security updates across government hosted servers and applications, is inconsistently practiced across agencies, with patching cadence and responsibility varying depending on whether a given system is hosted at the GIDC, on an agency's own premises, or with an external vendor, producing an uneven and difficult to audit security posture across the government's overall system estate.

7.3 Human Capital and Brain Drain

Civil service salary structures for ICT positions remain substantially below prevailing private sector compensation for comparable software engineering, systems administration, and cybersecurity roles, a disparity further amplified by the emigration of skilled ICT professionals to international labor markets. This compensation gap contributes to high turnover among trained government ICT staff, chronic understaffing of technical units within ministries and departments, and a resulting structural dependency on external vendors for both initial system development and ongoing maintenance, a dependency that in turn increases long run maintenance costs and creates

vendor lock in risk when proprietary systems are not accompanied by adequate source code escrow or technical documentation transfer to government custody.

7.4 Data Sovereignty and Cross Border Cloud Hosting Risks

As individual government agencies and donor funded digital projects increasingly evaluate commercial cloud hosting as an alternative to the capacity constrained GIDC, questions of data sovereignty, meaning the jurisdictional and legal control over citizen data hosted outside Nepal's borders, have gained prominence. In the absence of a finalized data localization policy or the enactment of the pending Information Technology and Cybersecurity Bill, decisions regarding whether particular categories of government data, especially biometric identity data and other sensitive personal information, may be hosted on foreign jurisdiction cloud infrastructure are made on an ad hoc, project by project basis rather than pursuant to a uniform, legally binding national policy.

Table 6: Risk Categorization, Operational Impact, and Mitigation Framework

Risk Category	Operational Impact	Current Mitigation Status	Recommended Mitigation
DDoS attacks on .gov.np domains	Service unavailability for citizen facing platforms	Ad hoc, reactive response; no unified DDoS scrubbing service	Deploy a shared, centrally managed DDoS mitigation layer for GIDC hosted domains
Website defacement and unauthorized access	Reputational damage; potential data integrity compromise	Incident specific remediation after the fact	Mandatory baseline hardening standards and periodic penetration testing
Absence of centralized SOC	Delayed detection of ongoing intrusions across the government estate	Fragmented, agency level monitoring where it exists at all	Establish a centrally funded, continuously staffed government SOC
Inconsistent patch management	Elevated exposure to known, publicly disclosed vulnerabilities	Agency discretionary patching cadence	Enforced patch management SLA tied to GIDC hosting agreements
ICT human capital attrition	Loss of institutional knowledge; heightened vendor dependency	Ad hoc retention incentives in limited units	Structured ICT career track with competitive allowances
Cross border cloud data sovereignty ambiguity	Uncertain legal jurisdiction over sensitive citizen data	No uniform data localization policy currently enacted	Statutory data localization requirement for defined sensitive data categories

CHAPTER 8: EMERGING TRENDS AND FUTURE OUTLOOK

Nepal's digital governance policy discourse has begun to reference a successor Digital Nepal Framework, generally referred to as Digital Nepal Framework 2.0, intended to update sectoral priorities and indicators in light of implementation experience since 2019 and the evolving technology landscape, including the growing policy relevance of artificial intelligence, cloud infrastructure sovereignty, and stronger cybersecurity governance. While the precise content and formal endorsement status of a successor framework continues to evolve and should be verified against the latest MoCIT publications at the time of reference, the direction of policy discussion signals an intention to move beyond the original framework's foundational digitization emphasis toward more sophisticated capabilities.

Artificial intelligence and predictive analytics are increasingly discussed as tools for public service dispatch optimization, including the potential application of automated triage to municipal grievance handling systems and analytics assisted detection of anomalous patterns indicative of tax fraud within the Integrated Tax System's transaction data. Realizing these applications at scale will require substantially improved data quality and interoperability, since machine learning based analytics are only as reliable as the underlying data pipelines feeding them, a precondition that, per the interoperability findings of Chapter 6, has not yet been achieved across most Nepali government systems.

Sovereign cloud infrastructure, referring to a nationally controlled cloud computing platform architected to host sensitive government workloads within domestic jurisdiction while offering elastic, commercially competitive provisioning characteristics, has emerged as a policy option under discussion as a potential successor or complement to the GIDC's current virtualization model. Complementary to this, edge data center concepts, involving smaller scale computing and storage capacity deployed at the provincial level rather than solely centralized in Kathmandu, have been proposed as a mechanism to reduce latency for provincial and local government systems and to improve resilience against single point of failure risk associated with total dependence on the Kathmandu based GIDC. Zero trust security architecture, which replaces the assumption of implicit trust within a network perimeter with continuous verification of every access request regardless of its origin, represents a further emerging concept relevant to strengthening the security

posture discussed in Chapter 7, though its adoption within Nepali government systems remains at an early, largely conceptual stage rather than an operationally deployed one.

CHAPTER 9: CONCLUSION AND STRATEGIC SUMMARY

This case study has traced Nepal's e governance trajectory from an early, predominantly informational web presence toward a partially transactional model characterized by flagship platforms such as the Nagarik App, the Integrated Tax System, and the e GP portal, situated within a legal and institutional architecture whose foundational statute, the Electronic Transactions Act 2063, predates the technological realities it must now govern. The country's improvement in the UN E Government Development Index, rising from 125th in 2022 to 119th in 2024 with a score of 0.5781, is a meaningful, empirically documented signal of progress, yet it should not obscure the structural weaknesses documented throughout this report: a shared infrastructure layer that falls short of internationally recognized data center redundancy standards, an interoperability framework that remains voluntary rather than enforced, a cybersecurity governance regime without a fully operational centralized Security Operations Center, and a digital divide between Bagmati Province and the Karnali and Sudurpashchim provinces that risks deepening rather than narrowing absent targeted intervention.

The overarching conclusion of this study is that Nepal's continued digital governance advancement is contingent less on the introduction of additional standalone applications than on three cross cutting, foundational imperatives. First, sustained political commitment across successive governments is required to prevent the policy discontinuity that has historically accompanied changes in ministerial leadership at MoCIT. Second, institutional interoperability, achieved through the enforced rather than voluntary adoption of the Nepal e Governance Interoperability Framework, must be prioritized as a precondition for realizing the full value of existing platforms such as the National ID and the Nagarik App. Third, legal modernization, most urgently the enactment of the long pending Information Technology and Cybersecurity Bill, is required to provide an updated statutory foundation capable of addressing cybersecurity governance, data localization, and platform accountability in a manner consistent with the technological realities of the current decade.

CHAPTER 10: ACTIONABLE RECOMMENDATIONS

10.1 For the Federal Government and MoCIT

- Prioritize the enactment of the pending Information Technology and Cybersecurity Bill within the current legislative session, with explicit provisions for data localization of sensitive citizen data categories and an updated cybercrime and platform accountability regime.
- Establish a dedicated, ring fenced budgetary allocation for Digital Nepal Framework implementation that is insulated from annual line ministry budget renegotiation, thereby reducing the risk of sectoral implementation stagnation observed in the Agriculture and Energy sectors.
- Mandate, through binding directive rather than voluntary guidance, the adoption of the Nepal e Governance Interoperability Framework for all new government system procurements, with non compliant procurements subject to review by the High Level e Governance Board.
- Strengthen the enforcement mechanism of the Individual Privacy Act 2075 by establishing a dedicated data protection oversight function with investigatory authority over both public and private data processors handling citizen data.

10.2 For Infrastructure Operators (NITC and NTA)

- Upgrade the Government Integrated Data Center's power redundancy, cooling capacity, and virtualization layer toward Tier III equivalent standards, and extend real time or near real time data replication to the Hetauda disaster recovery site for all classified critical government systems.
- Accelerate Rural Telecommunication Development Fund disbursement for last mile fiber connectivity projects in Karnali and Sudurpashchim provinces, with explicit multi year targets tied to measurable broadband penetration outcomes.
- Establish a centrally funded, continuously staffed Security Operations Center with monitoring authority across all GIDC hosted government systems, complemented by an enforced patch management service level agreement.

10.3 For Provincial and Local Governments

- Adopt a standardized, centrally certified municipal software suite for building permits, revenue management, and civil registration functions, replacing the current fragmented, vendor by vendor procurement pattern at the local government tier.
- Invest in structured digital literacy and technical capacity building programs for ward level and municipal civil service staff, with particular prioritization of rural municipalities in Karnali and Sudurpashchim provinces.

10.4 For Academia and the Domestic ICT Industry

- Align university curricula, including at institutions such as Himalaya College of Engineering, with the technical competencies required for government digital transformation, encompassing enterprise architecture, secure API design, and public sector data governance.
- Formalize structured public private partnership mechanisms that enable domestic ICT firms to participate in long term government system maintenance contracts with adequate source code and documentation transfer provisions, reducing vendor lock in risk while building domestic technical capacity.

Table 7: Prioritized Action Plan with Implementation Horizons

Recommendation	Responsible Stakeholder	Implementation Horizon	Expected Outcome
Enact the Information Technology and Cybersecurity Bill	Federal Parliament, MoCIT	Short term	Updated statutory basis for cybersecurity and data governance
Ring fenced DNF budgetary allocation	Ministry of Finance, MoCIT	Short term	Reduced sectoral implementation stagnation
Mandatory NeGIF adoption directive	High Level e Governance Board	Medium term	Reduced departmental data siloing
GIDC redundancy and DR replication upgrade	NITC	Medium term	Improved uptime for citizen facing platforms during peak demand
RTDF accelerated rural fiber disbursement	NTA	Medium term	Narrowed Karnali and Sudurpashchim connectivity gap
Centralized government SOC establishment	NITC, MoCIT, national CERT function	Medium to long term	Reduced detection and response time for cybersecurity incidents

Recommendation	Responsible Stakeholder	Implementation Horizon	Expected Outcome
Standardized municipal software suite	Local governments, MoCIT	Long term	Consistent local government digital service maturity nationwide
ICT curriculum alignment and PPP formalization	Academia, domestic ICT industry, MoCIT	Long term	Strengthened domestic technical capacity and reduced vendor lock in

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